Principal *Web3* Mobile & *Wallet* Engineer — *Rust*, Kotlin & Swift

Location: Remote / Poland (International Team)
Employment Type: Full-time / Part-time / Contract
Department: Mobile Engineering & *Blockchain* Infrastructure
Reports To: Head of Engineering / CTO
Seniority: Principal Engineer

About the Company

We are preparing to establish an independent software development company in Poland, focused on building modern financial technology, payment infrastructure, and *blockchain*-powered applications.

Our planned products and technical initiatives include:

- **Fintech & Payment Infrastructure:** Secure payment processing, financial transaction management, banking integrations, and digital payment services.
- ***Web3 Wallet*s & Digital Assets:** Mobile *cryptocurrency wallet*s, digital asset management, *wallet* connectivity, and *blockchain* transaction services.
- **Decentralized Finance (*DeFi*):** *Blockchain* integrations, decentralized exchange (*DEX*) connectivity, *token swap*s, staking, and cross-chain financial applications.
- **Digital Asset *Trading*:** Applications supporting *cryptocurrency* transactions, *trading* interfaces, and digital asset settlement.
- **Real Estate Investment Platforms:** Digital marketplaces and investment applications with selective *blockchain* integration.

Our goal is to combine the reliability and usability of traditional financial software with the capabilities of modern *blockchain* technology.

We are building an international engineering team and looking for experienced engineers who can help shape our technical architecture, establish engineering standards, and develop secure, scalable products from an early stage.

About the Role

We are looking for an experienced **Principal *Web3* Mobile & *Wallet* Engineer** to lead the technical architecture and development of mobile applications and *wallet* infrastructure across iOS and Android.

This is a senior hands-on engineering position combining native mobile development, *blockchain wallet* integration, *Rust*-based shared components, and technical leadership.

You will help *defi*ne how our mobile applications interact securely with *blockchain* networks, manage digital asset transactions, and deliver reliable financial experiences to users.

The ideal candidate has s*tron*g experience with ***Rust*, Kotlin, and Swift**, along with a deep understanding of *wallet* architecture, *crypto*graphic key management, *blockchain* transactions, and mobile application security.

We are particularly interested in engineers who have built or maintained production *wallet* applications, *cryptocurrency* financial products, or security-sensitive mobile platforms.

This role offers the opportunity to influence major architectural decisions while remaining closely involved in engineering implementation.

What You'll Do

Mobile Application Architecture & Engineering

- Design and guide the architecture of native Android and iOS financial applications.
- Develop and maintain mobile applications using Kotlin and Swift.
- *Defi*ne scalable, modular, and maintainable application architectures.
- Build shared components and reusable libraries where appropriate.
- Work closely with product designers and engineers to create secure and intuitive user experiences.
- Optimize application performance, startup time, memory usage, and responsiveness.
- Establish engineering standards for mobile code quality, testing, and release processes.
- Support application distribution, production monitoring, and mobile platform compatibility.

Web3 Wallet Engineering

- Design and implement *wallet* functionality for digital asset applications.
- Develop secure *wallet* creation, import, recovery, and account management workflows.
- Integrate *blockchain* networks, RPC providers, and *wallet*-related SDKs.
- Implement transaction preparation, signing, broadcasting, and confirmation tracking.
- Support *cryptocurrency* transfers, *token* balances, transaction histories, and *wallet* activity.

- Develop integrations for decentralized applications and *wallet* connectivity.
- Collaborate with *blockchain* engineers to support *DeFi* interactions, *token swap*s, and staking features.
- Evaluate *blockchain* network differences and *defi*ne consistent mobile *wallet* experiences across supported ecosystems.

Rust-Based *Wallet* Infrastructure

- Design and develop shared *wallet* libraries and *blockchain*-related components using *Rust*.
- Build reusable transaction processing, signing, serialization, and *crypto*graphic interfaces.
- Integrate *Rust* libraries into Android and iOS applications through appropriate FFI or language-binding mechanisms.
- Help *defi*ne clear interfaces between native mobile code and shared *Rust* components.
- Optimize performance and memory safety in security-sensitive functionality.
- Develop robust error handling, testing, and compatibility strategies for cross-platform libraries.
- Evaluate when shared *Rust* implementations offer practical advantages over native platform implementations.
- Contribute to the long-term maintainability and security of the shared *wallet* infrastructure.

*Crypto*graphy, Key Management & Mobile Security

- Design secure handling of private keys, recovery phrases, and signing operations.
- Implement secure storage using platform capabilities such as Android Keystore and iOS Keychain or Secure Enclave where appropriate.
- Apply *crypto*graphic standards and secure coding principles to *wallet* functionality.
- Design transaction authorization and user confirmation flows.
- Work with security engineers to review sensitive *wallet* components and integration patterns.
- Evaluate hardware-backed security, biometric authentication, and device-level protections.
- Protect applications against common mobile security risks, including sensitive data exposure and insecure communication.
- Support security testing, threat modeling, vulnerability remediation, and external audits.
- Help establish recovery and incident-response considerations for *wallet*-related functionality.

Blockchain & Digital Asset Integrations

- Collaborate on *wallet* support for *EVM*-compatible chains and other *blockchain* ecosystems.

- Integrate transaction and asset data from *blockchain* nodes, in*dex*ers, and service providers.
- Support *token* standards and contract interactions relevant to *wallet* applications.
- Design reliable transaction-state handling across pending, confirmed, failed, and replaced transactions.
- Consider network fees, nonce management, transaction simulation, and chain-specific confirmation requirements.
- Support integrations with *DEX* aggregators, payment services, and cross-chain providers.
- Work with backend engineers on transaction history, account synchronization, notifications, and monitoring.
- Help ensure consistency between mobile interfaces, backend transaction state, and *blockchain* data.

Technical Leadership & Architecture

- Act as a principal technical contributor for mobile *wallet* engineering.
- *Defi*ne architecture principles and guide important engineering decisions.
- Review critical code, technical proposals, and security-sensitive implementations.
- Mentor senior and mid-level mobile engineers.
- Collaborate with *Rust*, *blockchain*, backend, DevOps, and product teams.
- Evaluate technologies and architectural trade-offs based on product requirements.
- Help resolve complex technical problems involving *blockchain* integration, *crypto*graphy, and mobile platforms.
- Establish practical development, testing, and documentation standards.
- Remain hands-on with technically challenging components and proof-of-concept implementations.

Required Skills & Experience

- 8+ years of professional software engineering experience, with substantial mobile engineering experience.
- Proven experience designing and delivering production Android and iOS applications.
- S*tron*g proficiency in Kotlin and Swift.
- Practical experience developing production software or substantial reusable components using *Rust*.
- Experience integrating native mobile applications with shared libraries through FFI or language bindings.
- S*tron*g understanding of mobile application architecture, concurrency, asynchronous processing, and performance optimization.
- Experience building *Web3 wallet*s, *cryptocurrency* applications, or *blockchain*-connected financial products.
- Understanding of public/private key *crypto*graphy, digital signatures, *wallet* derivation, and transaction signing.

- Familiarity with *blockchain* RPC interfaces, transaction lifecycles, and *wallet* integration patterns.
- Experience implementing secure local storage, authentication, and transaction authorization.
- Knowledge of mobile security practices and protecting sensitive financial information.
- Experience developing and maintaining SDKs, libraries, or reusable engineering components.
- S*trong* automated testing, debugging, and production troubleshooting skills.
- Experience making architectural decisions and guiding engineering teams.
- Ability to communicate complex technical concepts and collaborate effectively across disciplines.
- S*trong* technical judgment and a practical approach to engineering reliability and security.

Nice to Have (Differentiators)

- Experience with production *cryptocurrency wallet*s or digital asset custody applications.
- Contributions to open-source *wallet* SDKs, *blockchain* libraries, or *Rust*-based infrastructure.
- Experience with *Wallet*Connect and decentralized application connectivity.
- Knowledge of *Ethereum*, *EVM* transaction formats, *Solana*, *Bitcoin*, or other *blockchain* ecosystems.
- Understanding of HD *wallet*s, BIP-32, BIP-39, BIP-44, and related *wallet* standards.
- Experience with account abstraction, smart accounts, multisignature *wallet*s, or MPC-based signing.
- Familiarity with ERC-4337, EIP-712 typed-data signing, or secure transaction simulation.
- Experience building *wallet* libraries using UniFFI, JNI, Swift interoperability, or comparable approaches.
- Knowledge of hardware *wallet*s, secure elements, or *trust*ed execution environments.
- Experience integrating decentralized exchanges, staking protocols, and *token swap* providers.
- Familiarity with mobile fintech, digital payments, or banking applications.
- Experience with secure software supply chains, dependency management, and reproducible builds.
- Experience mentoring engineers or leading cross-platform mobile architecture initiatives.

Education

- Bachelor's or Master's degree in Computer Science, Software Engineering, Computer Engineering, Mathematics, or a related discipline.
- Equivalent professional experience and substantial demonstrated engineering contributions are equally valued.

What Success Looks Like

First 3 Months

- Understand the company's planned *wallet*, payment, and mobile product requirements.
- Review the proposed mobile and *blockchain* architecture.
- Identify opportunities for shared *Rust* components and native platform integration.
- Establish initial security and engineering standards.
- Collaborate on prototypes for *wallet* connectivity, key management, and transaction workflows.
- Identify significant technical risks and architectural dependencies.

Within 6 Months

- Guide the development of foundational Android and iOS *wallet* functionality.
- Establish reliable integration patterns between *Rust*, Kotlin, and Swift components.
- Support secure transaction signing and *blockchain* network integrations.
- Improve mobile application architecture, testing, and release reliability.
- Mentor engineers and support technical design decisions across the mobile team.

Long-Term

- Help build a secure and scalable mobile *wallet* platform supporting the company's financial applications.
- Establish reusable *wallet* and *blockchain* components across mobile products.
- Improve security, reliability, and maintainability as the platform evolves.
- Guide mobile engineering standards and technical architecture.
- Support the expansion of digital asset, payment, and *DeFi*-related application capabilities.

Why Join Us?

- Help shape mobile *wallet* architecture at an early-stage independent software company.
- Work across *Rust*, Kotlin, Swift, *crypto*graphy, *blockchain*, and financial technology.
- Take ownership of important technical decisions while remaining actively involved in engineering.
- Contribute to applications involving digital asset payments, *wallet* infrastructure, and decentralized finance.
- Collaborate with an international engineering team.
- Work in a flexible, remote-friendly environment.
- Explore full-time, part-time, or contract collaboration arrangements.
- Help develop foundational technology that can support multiple financial products.

Ideal Candidate

We are looking for a highly experienced engineer who combines deep native mobile development expertise with a s*tron*g understanding of *blockchain wallet*s and secure systems.

You should be comfortable working across Kotlin, Swift, and *Rust*, designing reusable *wallet* components, and making architectural decisions that balance security, performance, and maintainability.

We value engineers who have solved difficult production problems, built reliable financial or *blockchain* applications, and helped other engineers improve their technical capabilities.

The ideal candidate is not only an experienced mobile developer but also a systems-oriented engineer who understands how *crypto*graphy, *blockchain* transactions, application architecture, and user experience fit toge*th*er.

Our ideal Principal *Web3* Mobile & *Wallet* Engineer combines hands-on technical depth, security-conscious architecture, and s*tron*g engineering leadership to help build reliable digital asset applications across Android and iOS.